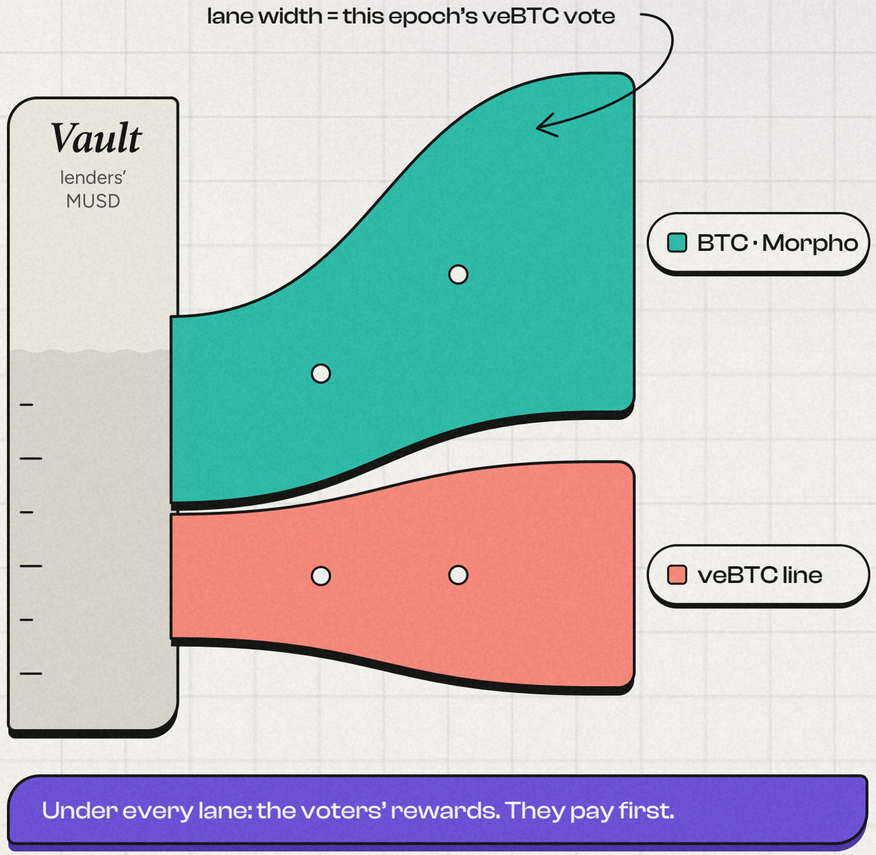
Surety

Lending gauges for Mezo

# veBTC holders decide where MUSD credit goes, and their rewards take *the first loss*.

One MUSD vault, steered by veBTC votes. The voters' own rewards stand in front of every credit line they back.




Live on Mezo testnet

`surety-credit.vercel.app`

Yehor Ivashchenko

Surety

The problem

# Plenty of BTC. Almost no *credit*.

**804** BTC

### locked in veBTC

Held by 1,796 holders. None of it can be borrowed against without unlocking. Mezo only does that for institutions, through Mezo Prime.

**1**

### open trove

On Mezo's public MUSD CDP at block 12,705,315. Redemptions eat troves while MUSD trades about 1% under $1.

**0**

### loss for whoever picks

In money markets today a curator or governance decides where deposits go. When the pick goes bad, it carries none of the loss.

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From Mezo's blog · "Aerodrome for lending"

# "...the system cannot tell who is serious"

mezo.org/blog/aerodrome-for-lending

WHAT MEZO ASKED FOR

Lending gauges: veBTC votes steer capital to lending markets, as a risk signal.

WHAT IT LEFT OPEN

The mechanism. It is saved for "a future post".

SURETY'S ANSWER

Make the vote cost something. The voters' rewards take the first loss.

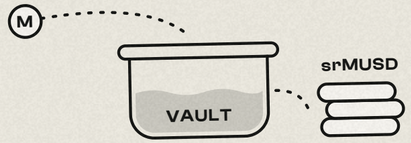
Surety

How it works

# Four verbs, *one* vault.

Lenders bring the MUSD. veBTC holders steer it and stand behind their picks. Borrowers draw it with no redemption risk.

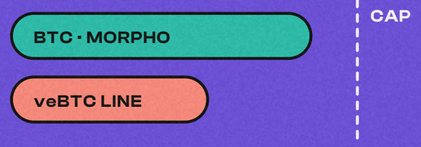
01 for lenders




## Lend.

Deposit MUSD into a Morpho VaultV2 vault. You hold srMUSD shares.

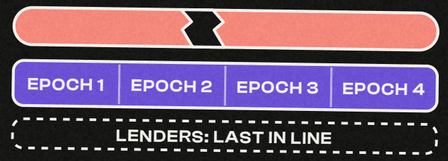
02 for veBTC holders




## Vote.

veBTC power is snapshotted at epoch start. The allocator moves capital to match the votes, inside vault caps.

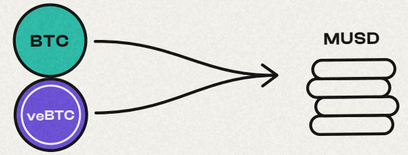
03 for the voters, again




## Underwrite.

Voters earn 10% of interest plus sponsor incentives. Those rewards season 4 epochs and pay first if a line they backed goes bad.

04 for borrowers




## Borrow.

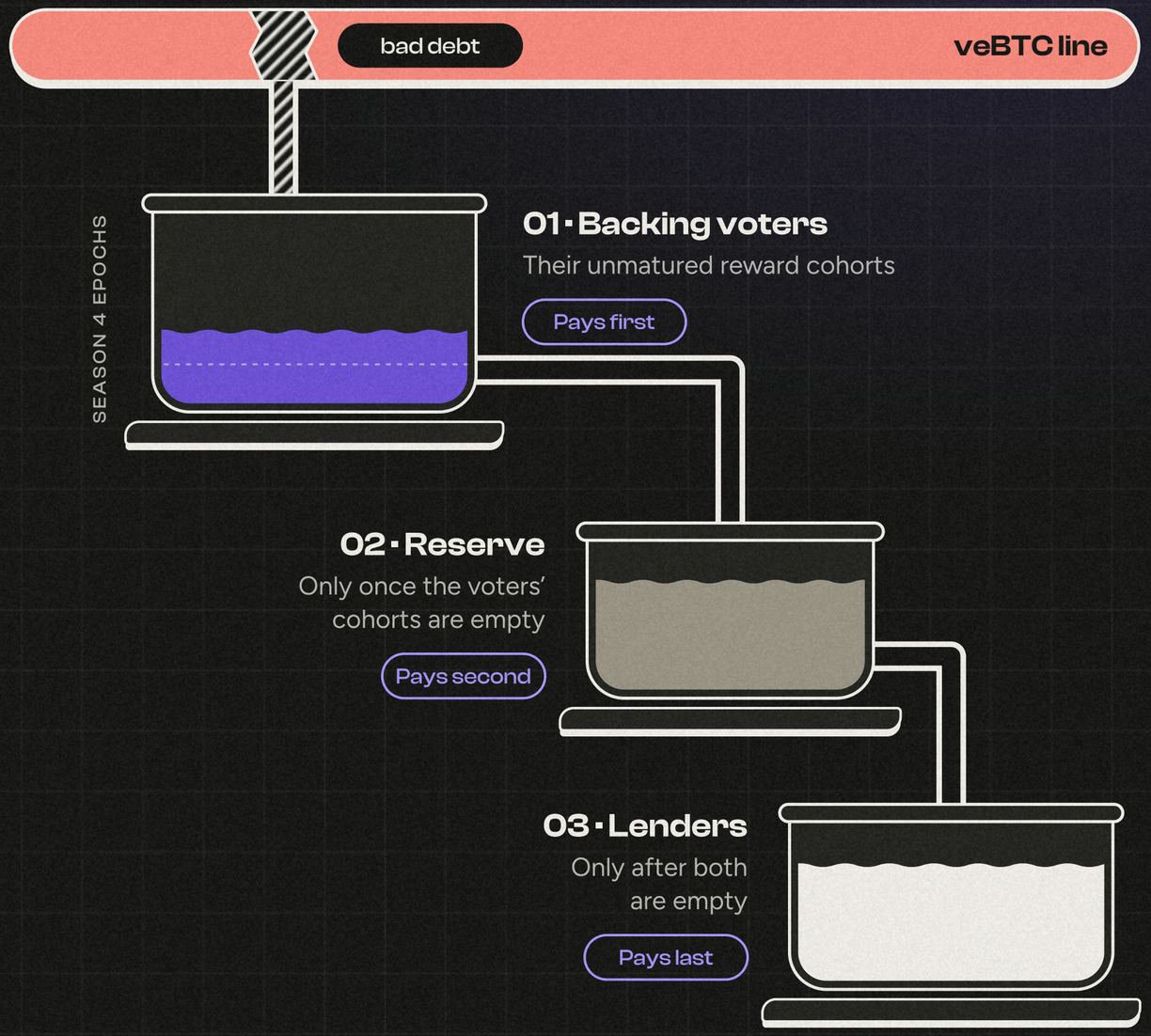
MUSD with no redemption risk: against BTC on Mezo's Morpho Blue, or against a locked veBTC that keeps voting.

Surety

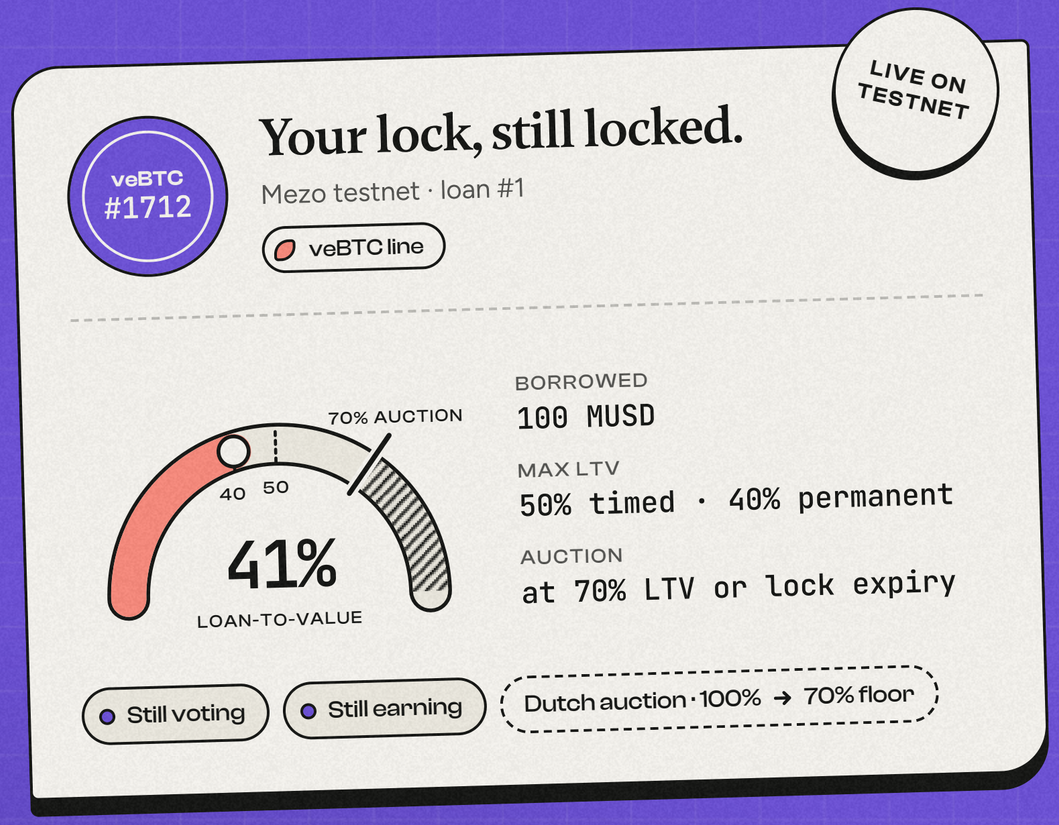
Who pays first

# When a loan goes bad, the voters pay first.

| | |
|---|---|
| **Order** | Backing voters' unmatured cohorts, then the reserve, then lenders. |
| **Size** | Cover is sized so lenders return to the pre-loss share price. |
| **Gate** | New deposits pause while a cover is re-recognised, so nobody can skim it. |



Surety




*Mezo Prime does this for institutions. Surety does it for any lock.*

For veBTC holders

# Borrow against your veBTC without unlocking it.

| | |
|---|---|
| **Stays locked** | Each loan gets its own escrow. The lock keeps voting and earning on Mezo. |
| **Borrow up to** | 50% LTV on a timed lock, 40% on a permanent one. |
| **If it slips** | At 70% LTV or at lock expiry, a Dutch auction runs from 100% of the lock's value down to a 70% floor. |
| **No redemptions** | The MUSD comes from the vault, so nobody can redeem against you. |

Surety

Built on Mezo

# Made from Mezo's *own* parts.

### MUSD 01

`lent · borrowed · incentives`

What lenders deposit, borrowers draw and sponsors post.

### veBTC 02

`votingPowerOfNFTAt`

Vote weight, snapshotted at the start of each epoch.

### Mezo voting + rewards 03

`PoolsVoter · RewardsDistributor`

Reached through each loan's escrow, so a pledged lock keeps voting and claiming.

### Skip BTC/USD 04

`precompile · fail-closed at 60 s`

Prices every loan and auction. A price older than 60 s stops them.

### Morpho Blue on Mezo 05

`LLTV 0.86 · AdaptiveCurveIrm`

The BTC market the vault supplies, where BTC holders borrow MUSD.

### VaultV2 + adapter 06

`MorphoMarketV1AdapterV2`

Morpho's vault and adapter, vendored unmodified at the exact commits Mezo runs.

GAUGE-READY

Mezo's own `ERC4626VaultAdapter` fits the vault. One governance call switches on MEZO emissions for lenders.

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Live on Mezo testnet

Current deployment · all 14 contracts Sourcify exact match · every step is a transaction

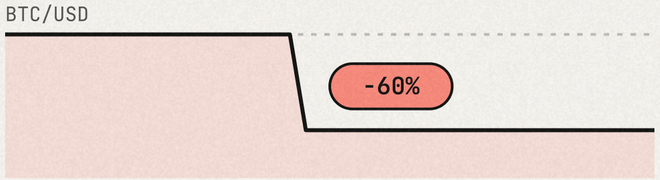
# One run, from deposit to bad debt.

## SETUP — lend, vote, borrow

| | | |
|---|---|---|
| 01 | Deposit<br>0xb0f5…1b7f | 1,000 MUSD |
| 02 | Vote · veBTC #1711<br>0x457d…3c19 | 70 / 30 |
| 03 | Vote · veBTC #1712<br>0xa237…2168 | 30 / 70 |
| 04 | Rebalance<br>0xea9d…d272 | 450 / 450 MUSD |
| 05 | Sponsor incentives<br>0x4028…0bd9 · 0xc9ad…a124 | 100 + 150 MUSD |
| 06 | veBTC loan<br>0x050d…9fe9 | 100 MUSD · 41% LTV |
| 07 | Morpho borrow<br>0xb8b2…b9d2 | 250 MUSD · 76.9% LTV |

## STRESS — break it on purpose

| | | |
|---|---|---|
| 08 | BTC price shock<br>0x1a75…0434 | -60% |
| 09 | Morpho liquidation<br>0x9fd3…0723 | 125.40 bad debt |
| 10 | veBTC auction<br>0x32c4…79ad | 68.24 vs debt 100 |




**veBTC shortfall** 31.76 MUSD

## COVER — voters first

| | | |
|---|---|---|
| 11 | Record loss · Morpho<br>0x34c0…00a8 | |
| 12 | Record loss · veBTC<br>0x9aeb…da00 | |
| 13 | Cover · Morpho line<br>0x7a52…8b03 | |
| 14 | Cover · veBTC line<br>0xf260…ae9b | |

**Deposits paused**
automatically while the covers re-recognise

Surety

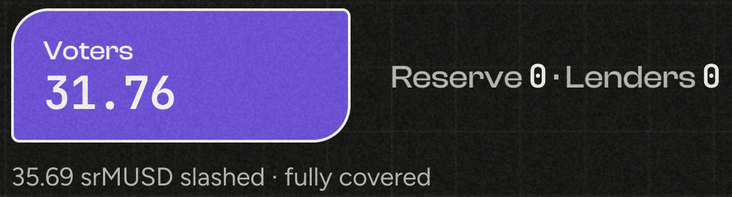
Who paid · same testnet run

# 157.16 MUSD of bad debt. Lenders bore *36.98*.

### veBTC line

**31.76 MUSD**

Auction at the floor: 68.24 bid vs 100 debt




### BTC · Morpho

**125.40 MUSD**

Liquidation left bad debt




**DEPOSIT GATE** New deposits paused automatically during re-recognition, so nobody could buy in cheap and skim the cover.

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Engineering & security

# Tested like it already holds *money*.

**197**

**Foundry tests**

Including 7 invariants at 256 runs × 64 depth.

**9**

**regression tests**

Written from an independent security review.

**0**

**critical or high**

The review found 4 medium and 6 low. All fixed.

**98**

**TypeScript tests**

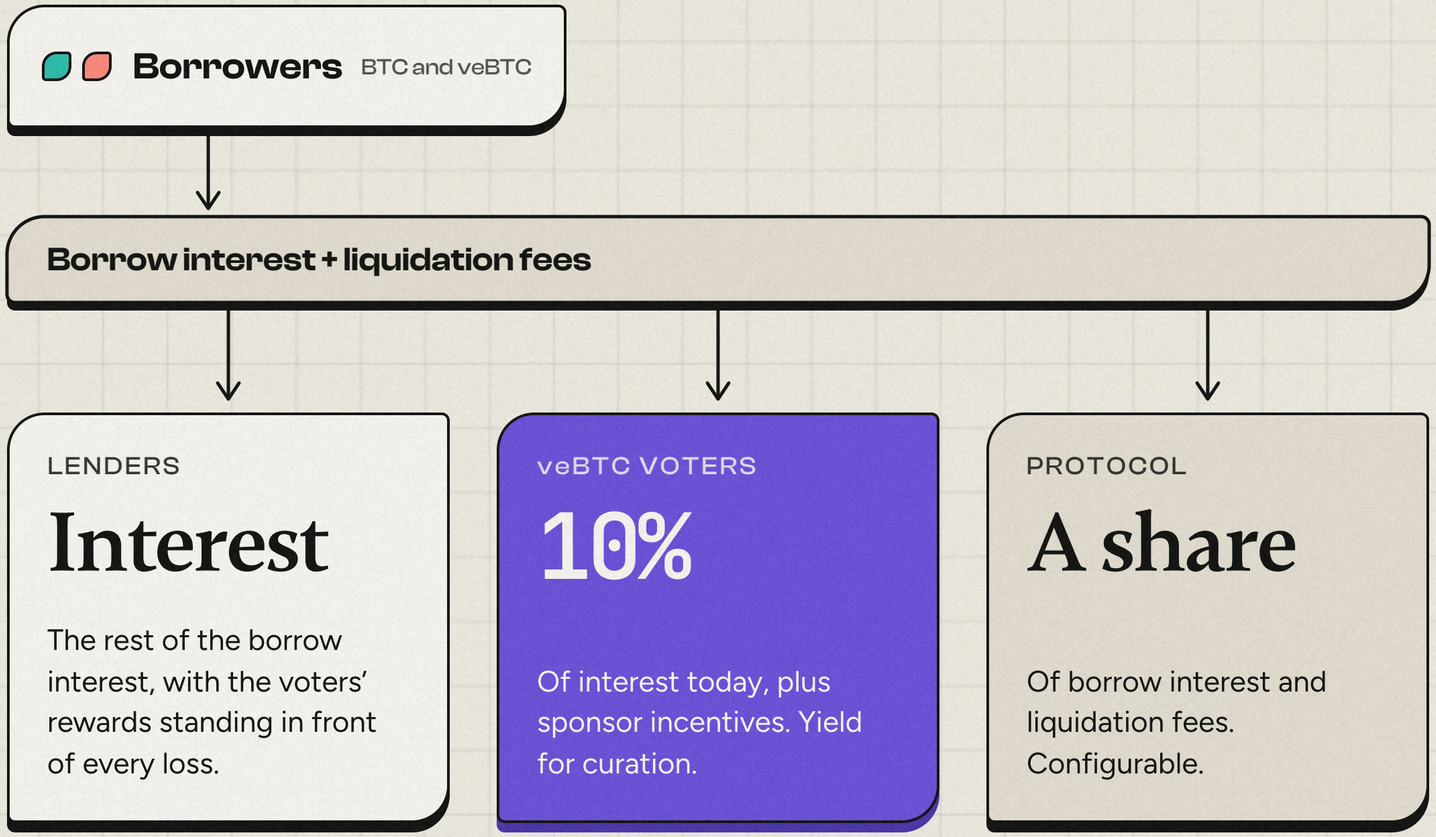
Across the SDK, the keeper and the MCP server.

SHIPS WITH

- Permissionless keeper
- TypeScript SDK
- MCP server for agents
- Web app · `surety-credit.vercel.app`

Business

# Borrowers pay for the whole loop.




WHAT MEZO GETS

01 Demand for MUSD.

02 A use for veBTC beyond fees.

03 The lending-gauge mechanism its blog asked for.

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Roadmap

# Capped first. Then *open it up*.

**NOW**

Live on Mezo testnet, 14 contracts verified

Mezo mainnet: capped launch

**NOV 2026 · WAVE 2**

Real Mezo vault gauge, through governance

MEZO incentives

Self-repaying veBTC loans

**DEC 2026**

Owner roles to timelock + multisig

External audit

**Q1 2027**

Raise caps

More BTC collateral lines

Surety

Try it

# Votes with something *on the line*.

LIVE APP

`surety-credit.vercel.app`

REPOSITORY

LINK PENDING

VIDEO

LINK PENDING

MAINNET

Mezo mainnet: capped launch

BUILT BY

Yehor Ivashchenko solo developer

X

`@BazydloMis57623`

# Surety